

and capacities in general. Faster bit rates and capability to test a larger number of devices in parallel are evident.

As an example, Jin Yokoyama, functional manager-memory test, Advantest, explains Advantest's test setup evolution. The latest device in their T5503 series has seen size reduction and improved cost efficiency with a capacity to test 512 DDR4 DRAM memory devices simultaneously. Their LPDDR testing capability has evolved from 1600 megabits per second (Mbps) and 12.8 gigabytes per second (GB/s) for LPDDR3 memories to 3200Mbps and 25.6GB/s respectively for the newer LPDDR4. On the other hand, the newest T5582 series can support 1536 devices under test (DUT) in parallel with maximum data rate of up to 1.2Gbps.

Higher capacities in high-bandwidth oscilloscopes. High-bandwidth oscilloscopes are used for parametric tests—evaluating the signal integrity of the memory. The upgraded oscilloscopes support mixed-signal channels, that is, these have four analogue channels and 16 digital channels. These channels help in a range of functions, like connecting the DDR command bus to evaluate command triggers or assessing the complete protocol of the command.

Another characteristic of DDR memory signals is that these come as a combination of 'read' and 'write' commands. 'Read' command comes from the memory itself, while 'write' command comes from the controller. Read-write channels of the oscilloscope help in separating these commands.

Bhatia explains, "With the introduction of DDR4, the oscilloscope bandwidth has gone up to 13GHz. The previous highest bandwidth range was 8GHz. The upgrade has resulted in a net 20 per cent increase in equipment price."

Quad-sampling in logic analysers. Functional testing of high-speed memories requires capturing of all possible channels at the same time. Logic analysers are essential devices for the purpose. Logic analysers suited for DDR testing have as many as 136 or more channels and a speed of 4Gbps or higher.

As a major upgrade, many manufacturers are developing logic analysers



Logic analyser for high-speed memory testing (Image source: eet.com)

with a feature called quad-sampling, which means these analysers can take four samples in every clock cycle. The DDR works on both rising and falling edges of a signal. Moreover, it has both read and write signals on each rise and each fall. For efficient testing, in one clock cycle, one sample each of Read rising (Read command on the rising edge of the signal), Write rising, Read falling and Write falling is needed.

Without quad-sampling technology, the tester will have to split the clock cycle into four samples by physical techniques and do the sampling on four different channels, which is very cumbersome. Quad-sampling allows you to procure all four samples on a single channel, cutting down the complexity, time and effort required.

Bhatia shares that Keysight's previous version of logic analysers used to have dual-sampling capability, driving testers to procure the samples on two separate channels. These have been upgraded to quad-sample now.

Additionally, the speed of analysers has gone up to 4Gbps from 2.5Gbps earlier. The protocol testing setup can be arranged in two levels: Either for only command and address debugging, or with data debug capability along with command and address debugging.

Software upgrade. There are specialised software to test DDR bus for compliance to JEDEC standards. These DDR compliance testing software are fully automated and can complete 50-60 tests in one go. Another software called DDR debug tool analyses failures detected by the DDR compliance testing software. It

helps to narrow down reasons behind the failure and locate the particular bus that has failed.

Tips for optimal T&M

Following the upgrade trends in DDR T&M devices, there will be certain criteria and challenges in designing and selecting the optimal test setup. Bhatia shares the important ones.

Preference to devices with the highest test coverage. DDR doesn't have a dedicated compliance program like other technologies. It has a set of specifications following which the tests are done. There is a large list of tests that can be conducted for DDR memories. Different vendors provide support for different number of test automation—some provide 40 to 50 tests, while others may give 100. Before selecting a setup, check the number of tests that it supports.

Don't over-invest in higher-bandwidth devices. Bhatia opines, "DDR does not require very high-bandwidth values such as 20GHz. So test equipment with 13GHz bandwidth will work fine. DDR memories use FR4, which is a low-cost semiconductor material. As these are made to keep the cost affordable, excessively high-level bandwidth is not achievable."

Data debug is a good choice. Some protocol testing equipment are incapable of data debugging. Logic analysers with data, address and command debug capability can be customised to debug any of the selected parameters as required. So go for a logic analyser with data debug capability.

Use RC-type probing for best results. For DDR probing, ensure use of resistor-capacitor (RC)-type probes. Many designers choose RCRC (high-bandwidth) type probes, which do not give good results because they do not generate high impedance.

Going forward

DDR5 is still under development, but is projected to become mainstream in a short span of time. So T&M equipment manufacturers are already upgrading for the future. For example, Advantest is improving its test devices for a massive 6.4Gbps speed to support LPDDR5. **EFY**